

Data Forge Challenge #002: E-Commerce Sales & Profitability Analysis

Project Overview

As part of Data Forge Challenge #002, I analysed an e-commerce sales dataset to understand overall business performance, product profitability, sales volume, and payment behaviour. The analysis focused not only on total sales, but also on identifying products and categories that contribute to profit and those that may be creating profitability problems.

I used SQL for data analysis and investigation, Power BI for visualization, and DAX for calculated measures and KPIs.

Tools Used

- SQL — data exploration, aggregation, profitability analysis and investigation of business questions
- Power BI — dashboard development and interactive data visualization
- DAX — KPI calculations and profitability measures

Business Questions

The analysis focused on questions such as:

- What are the overall sales and profitability performance?
- Which products generate the highest profit?
- Which products are contributing to losses?
- Which categories and sub-categories perform well or poorly?
- Which payment method is most frequently used?
- How do sales volume and revenue compare across categories?
- Are there products with high sales volume but negative profitability?
- How does payment method relate to transaction size and profitability?
- Which product sub-categories have the strongest and weakest profit margins?

Dashboard Structure

Page 1 — Executive & Performance Overview

The first page provides a high-level view of the business.

KPIs

- Total Sales: ₹431,771
- Total Profit: ₹36,963

- Total Quantity: ₹5,615
- Profit Margin: 8.44%

The page also includes:

- Most-used payment method
- Volume vs Revenue comparison
- Top Profit Winners matrix
- Bleeding Loss List matrix

The purpose of this page is to allow a business stakeholder to quickly understand overall performance and identify products requiring further investigation.

Page 2 — Profitability Deep Dive

The second page focuses more specifically on product and sub-category profitability.

It contains:

- Total Sales
- Total Profit
- Total Quantity
- Profit Margin
- Profit Margin by Sub-category
- Category slicer
- Sub-category slicer
- Payment Mode slicer
- Comparative analysis of selected products and payment methods

The page also highlights specific comparisons such as Handkerchief vs Printers and Credit Card vs UPI.

Key Insights

1. Overall profitability is relatively low compared with sales.

The business generated ₹431,771 in sales and ₹36,963 in profit, resulting in an overall profit margin of approximately 8.44%. This indicates that generating sales does not necessarily translate into strong profitability. Therefore, looking only at revenue would hide some of the underlying performance problems.

2. T-shirts had one of the strongest profit margins

The analysis showed that T-shirts had the highest profit margin at approximately 20.32%. This makes T-shirts an important product from a margin perspective, even though a high margin does not automatically mean the product generates the highest total profit.

This distinction is important:

Profit margin tells us how efficiently a product converts sales into profit, while total profit tells us how much profit the product contributes.

3. Printers were a major contributor to total profit

Printers generated approximately ₹8,606 in total profit, making them one of the strongest profit contributors identified in the analysis. However, the analysis also showed that their profit contribution needs to be considered alongside their revenue and margin rather than looking at profit alone. This is why the dashboard separates the Top Profit Winners from the overall KPI section.

4. Cash on Delivery was the most-used payment method

Cash on Delivery recorded 684 orders, making it the most frequently used payment method in the dataset. This provides a useful view of customer payment preferences.

However, payment frequency alone does not mean that Cash on Delivery is the most profitable payment method. Revenue, transaction size and profit need to be considered separately.

5. Some products are creating profitability problems

The loss analysis identified products with negative profit contributions, including products such as Kurti, Skirt and Leggings in the loss-producing group identified during the investigation. This is important because a product can still generate sales while reducing overall profitability.

The Bleeding Loss List was therefore included in the dashboard to make these products easier to identify.

6. High sales volume does not always mean high profitability

One of the more interesting findings was the performance of Furniture. Furniture recorded a relatively high sales volume of approximately 310 but generated negative net returns of around ₹806.

This demonstrates an important business lesson: Products with high demand are not necessarily profitable products. A business could therefore make the mistake of prioritizing a product simply because it sells frequently, while overlooking the fact that it is losing money.

7. Clothing generated the highest revenue among the categories examined

The category comparison showed Clothing generating approximately ₹144,323 in revenue, compared with approximately ₹166,267 for Electronics and ₹127,181 for Furniture. However, revenue performance did not automatically translate into equivalent profitability performance.

This reinforces the need to analyse revenue, volume, and profit together rather than relying on a single metric.

8. Payment method differences become more noticeable for individual products

The investigation into Handkerchief vs Printers and Credit Card vs UPI showed that payment behaviour can differ depending on the product. For example, the analysis noted differences between Credit Card and UPI transactions for expensive products such as Printers.

This led to a further analytical question around whether payment method and transaction size are related. Rather than claiming that one payment method causes higher or lower profit, this should be treated as a pattern worth investigating further.

Recommendations

1. Investigate products with negative profit

Products consistently generating negative returns should be investigated before increasing their sales volume. The business should review:

- Selling price
- Cost price
- Discounts
- Procurement costs
- Product-level margins
- Whether the product should remain in the current assortment

The goal should not simply be to sell more, but to determine whether the product can be made profitable.

2. Review high-volume but loss-making products

Furniture is a good example of why volume should not be used as the only performance indicator. Products with strong sales volume but negative profit should be prioritized for margin investigation. Possible actions include reviewing pricing, reducing costs, renegotiating supplier prices, or adjusting discounts.

3. Protect high-margin products

Products such as T-shirts, which demonstrate strong margins, should be monitored closely. The business could investigate whether these products can support:

- Higher sales volume

- Cross-selling
- Bundling
- Better inventory availability

However, this should be based on demand and total contribution, not margin percentage alone.

4. Evaluate payment methods using profitability metrics

Cash on Delivery is the most-used payment method, but usage frequency alone does not establish that it is the most financially beneficial option. The business should compare payment methods using:

- Number of transactions
- Revenue
- Average transaction value
- Profit
- Profit margin

This would provide a more complete picture of the financial impact of each payment method.

5. Monitor category performance using multiple metrics

Clothing generated the highest revenue among the categories examined, but management should avoid assuming that the highest revenue category is automatically the most profitable. Category performance should be monitored using:

Sales → Quantity → Profit → Profit Margin.

This makes it easier to identify categories that generate high sales but weak financial returns.

Key Business Takeaway

The biggest lesson from this analysis was that sales volume and revenue alone do not tell the full story of an e-commerce business.

A product can sell frequently and generate substantial revenue while still producing a negative return. At the same time, a product with a smaller sales contribution can have a much stronger profit margin.

Therefore, profitability analysis should consider revenue, volume, total profit, and profit margin together.

Challenge / What I Learned

Data Forge Challenge #002 pushed me beyond simply creating charts from a dataset.

The challenging part was moving from: “What happened?” to “Why is this happening, and what should the business investigate next?”

For example, identifying that Furniture had high sales volume but negative returns led to a deeper profitability question rather than stopping at the initial observation. The project also helped me practice turning SQL results into business insights and then communicating those insights through an interactive Power BI dashboard.

Final Dashboard

The project consists of a two-page Power BI dashboard:

Page 1 — Executive & Performance Overview

- Total Sales
- Total Profit
- Total Quantity
- Profit Margin
- Most-used Payment Method
- Volume vs Revenue
- Top Profit Winners
- Bleeding Loss List

Page 2 — Profitability Deep Dive

- KPI overview
- Profit Margin by Sub-category
- Category/S-category/Payment Mode slicers
- Handkerchief vs Printers comparison
- Credit Card vs UPI comparison
- Supporting profitability insights

Executive and Performance Overview

Total Sales

₹438K

Total Profit

₹37K

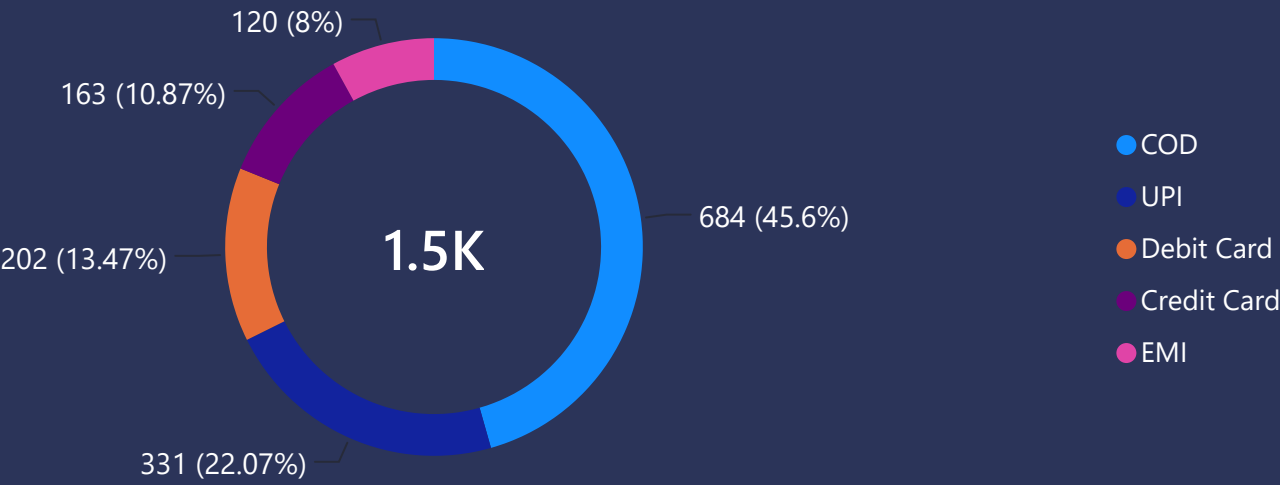
Total Quantity

₹6K

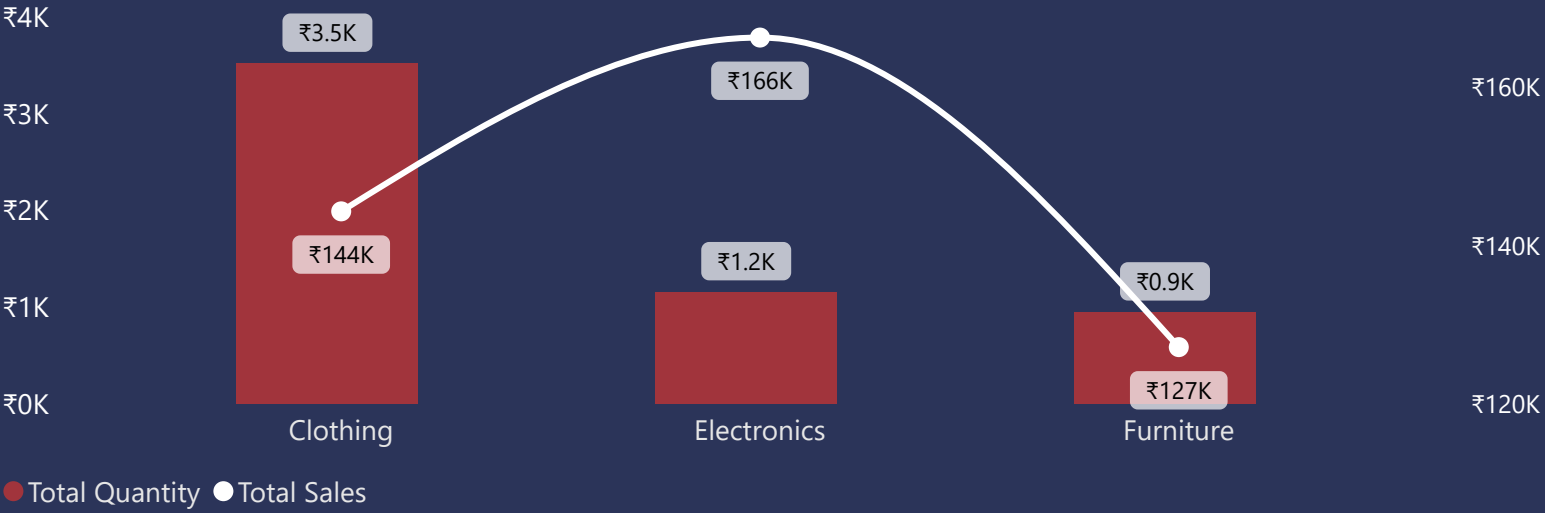
Profit Margin

8.44%

Most Used Payment



Volume vs Revenue



Top Profit Winners

sub_category	Total Profit	Profit Margin
Printers	₹8,606	14.52%
Bookcases	₹6,516	11.46%
Saree	₹4,057	6.87%
Accessories	₹3,353	15.43%
Tables	₹3,139	13.88%
Trousers	₹2,847	9.48%
Stole	₹2,431	13.11%
Phones	₹1,847	4.00%

The Bleeding Loss List

sub_category	Total Profit	Profit Margin
Furnishings	-₹806	-5.98%
Electronic Games	-₹644	-1.64%
Kurti	-₹401	-11.93%
Skirt	-₹315	-16.19%
Leggings	-₹130	-6.17%

Payment mode and Customer Behaviour

Total Sales

₹438K

Total Profit

₹37K

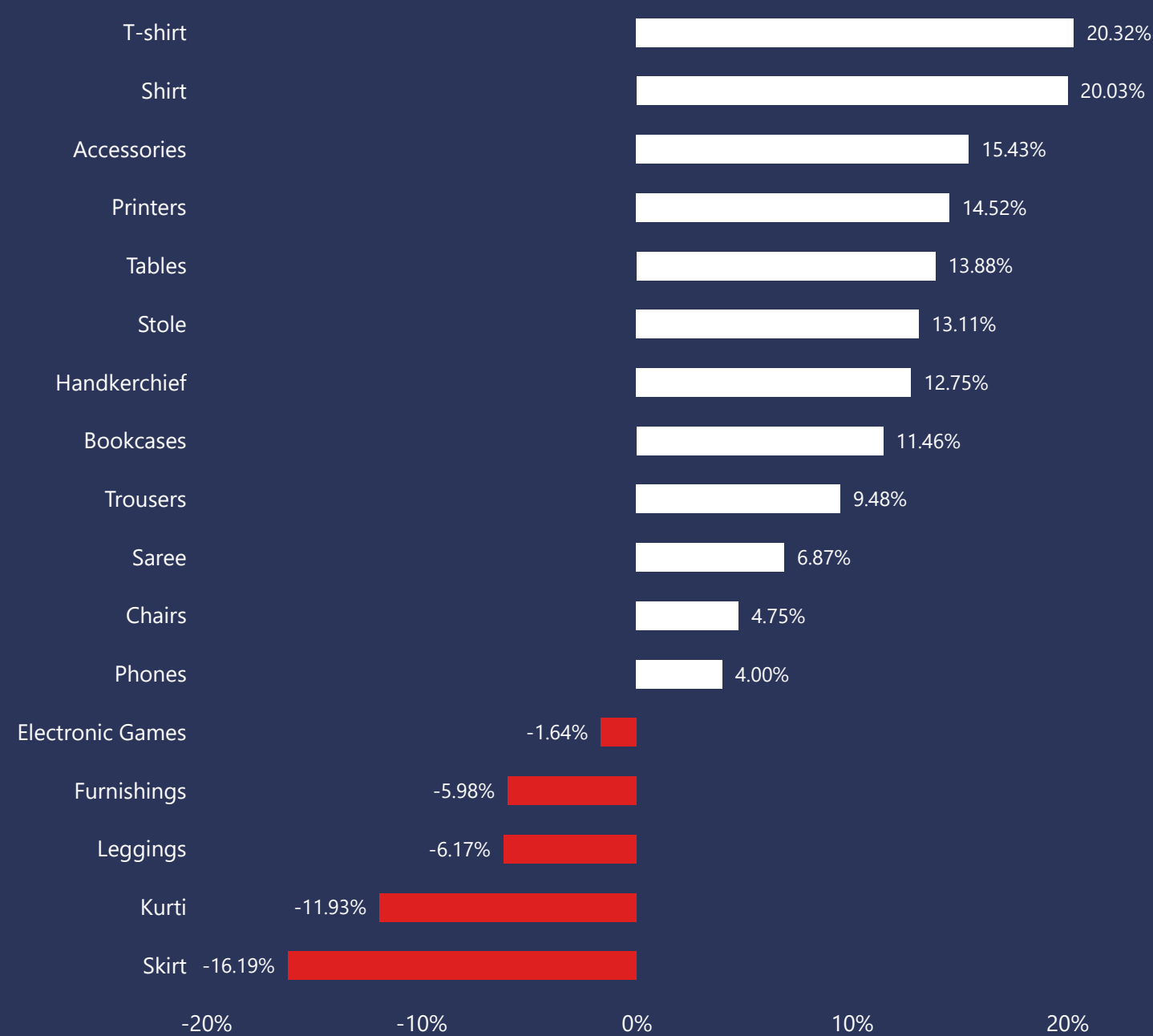
Total Quantity

₹6K

Profit Margin

8.44%

Profit Margin by sub_category



UPI dominates low-cost everyday retail (Handkerchief), driving over double the volume and capturing 588 INR in profit. However, for high-ticket electronics (Printers), UPI fails to protect margins and bleeds a net loss of -59 INR, where Credit Cards securely dominates profitably.

sub_category	Credit Card	UPI
Handkerchief		
Total Quantity	₹79	₹182
Total Sales	₹1,455	₹3,619
Total Profit	₹92	₹588
Printers		
Total Quantity	₹64	₹52
Total Sales	₹15,963	₹7,229
Total Profit	₹1,920	-₹59

Category

Clothing	Electro...	Furnitu...
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Sub_Category

Access...	Bookca...	Chairs
Electro...	Furnis...	Handk...
Kurti	Leggin...	Phones

Payment Mode

COD	Credit C...	Debit C...
EMI	UPI	